Intrusion Attack Detection

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1 Data source

https://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html

2 Task descriptions

This database contains a standard set of data to be audited, which includes a wide variety of intrusions simulated in a military network environment.

A connection is a sequence of TCP packets starting and ending at some well defined times, between which data flows to and from a source IP address to a target IP address under some well defined protocol. Each connection is labeled as either *normal*, or as an *attack*, with exactly one specific attack type.

Task:

To build a network intrusion detector, a predictive model capable of distinguishing between "bad" connections, called intrusions or attacks, and "good" normal connections.

3 Methodology

Choose one of the following methods:
Clustering algorithm: Apply one of the following approaches:

- k-means or
- DBSCAN

Classification algorithms: Apply one of the following approaches:

- Multilayer neural network (NN)
- Decision tree (DT)
- Ensemble learning

Implement the classification or clustering algorithm.
Tune the parameters to find the optimal configuration (eg. learning rate, number of epocs (NN), stoping criterions (DT), ...).

4 Testing model

- Train: kddcup.data_10_percent.gz
- Test Unlabeled: kddcup.testdata.unlabeled_10_percent.gz
- Test target labels: corrected.gz